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Introduction

This report examines U.S. international trade from the 2010s to present, using data from the Census Bureau API and FRED respectively. We explore two main questions: what does the U.S.trade and with whom, and what economic factors influence how much the U.S. exports?

To answer the first question, we analyze trade flows across industries and countries, looking at how patterns shifted before and after the disruptions of 2020. Understanding these structural trends provides a clearer picture of where the U.S. economy stands in the global trade landscape.

To answer the second question, we examine three economic factors that might drive export levels: exchange rates, oil prices, and interest rates. The factors are analyzed both visually and through correlation calculations to assess how strongly they relate to U.S. export performance.

Together, these two areas of analysis paint a broader picture of how U.S. trade is shaped not just by market forces, but by policy decisions, global events, and long-term structural shifts.

US Trade Overview & Analysis of the International Trade Landscape

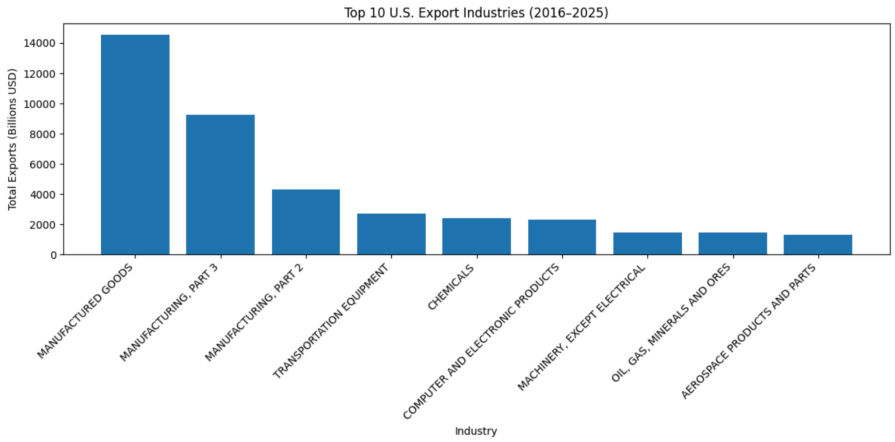
Census Bureau Data and API Insights (2016–2025)

To start, we explore the fundamental patterns of U.S. international trade using data accessed via the Census Bureau API from 2016 to 2025. By analyzing North American Industry Classification System (NAICS) codes and country-level end-use data, we can decide which industries and global partners drive the U.S. economy and how these dynamics shifted following the major structural disruptions of COVID.

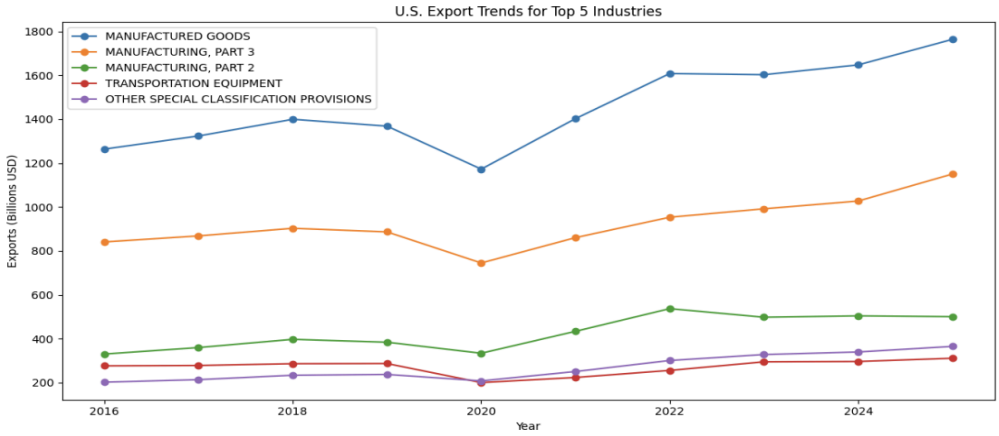
Top U.S. Export Industries

The U.S. export sector is dominated by manufacturing and industrial output. Manufactured Goods lead the rankings, followed by specialized sub-sectors like Manufacturing Part 3 and Manufacturing Part 2. Sectors that come after include transportation equipment, chemicals, and computer products (Fig. 1).

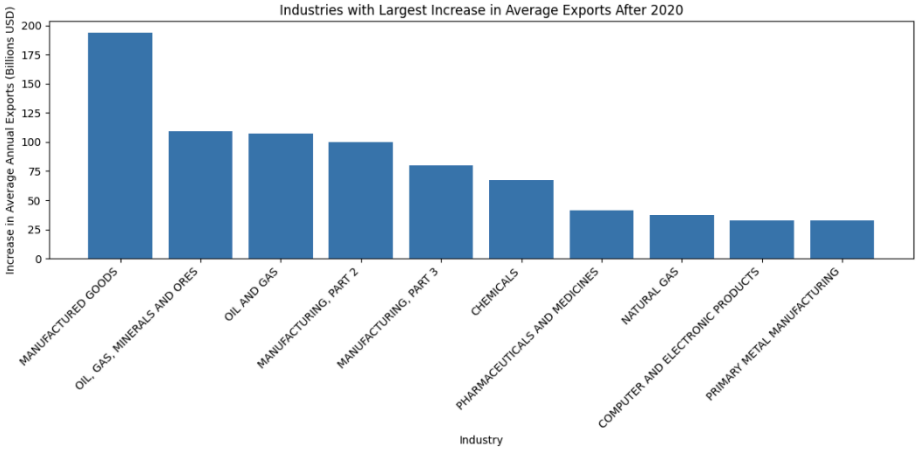
The data indicates that U.S. exports are highly concentrated in industrial sectors rather than primary commodities.The 2020 pandemic caused a sharp contraction across all top-five industries, while the recovery was notably rapid. By 2025, manufactured goods reached record-high values. This resilience suggests that global demand for U.S. technical expertise and industrial machinery remains a cornerstone of the international trade landscape shown by the high gradient from 2020 to 2022 (Fig. 2).



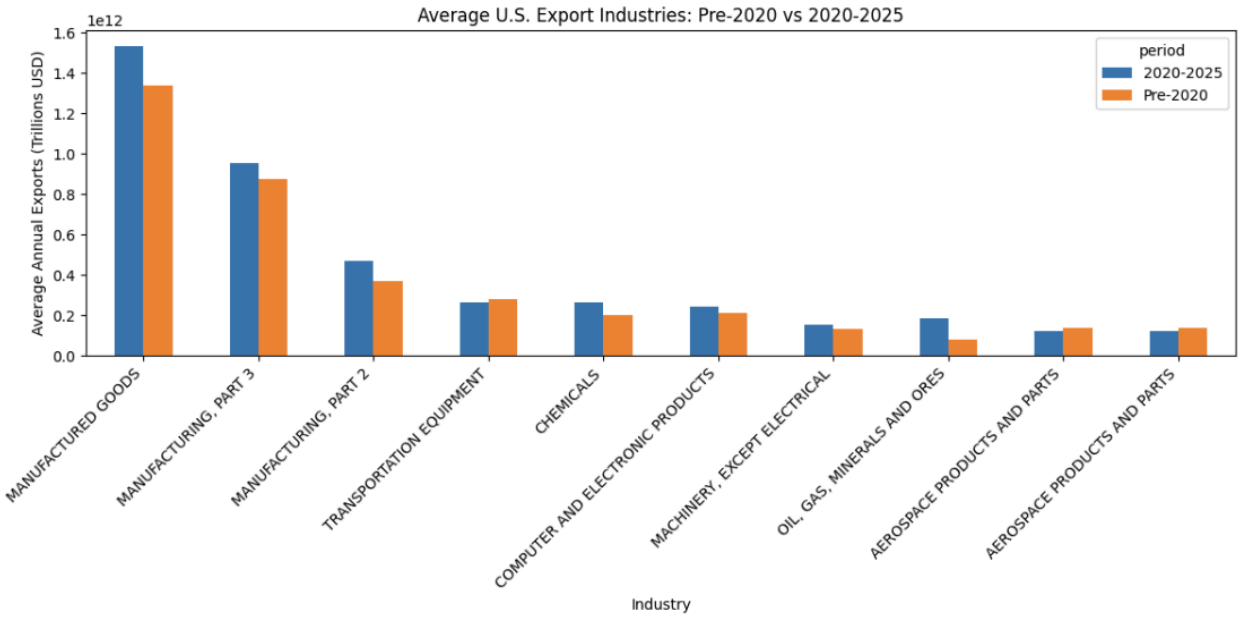
(**Fig. 1.** Top 10 U.S. Export Industries, 2016–2025)



(**Fig. 2.** U.S. export trends for the top five industries, 2016–2025)



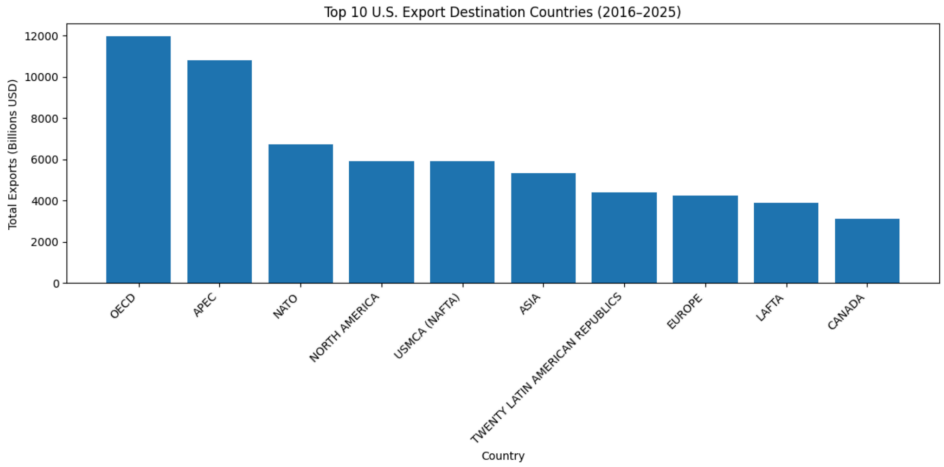
(**Fig. 3.** Industries with the largest increase in average U.S. exports after 2020)



(**Fig. 4.** Average U.S. export industries before 2020 and from 2020 to 2025)

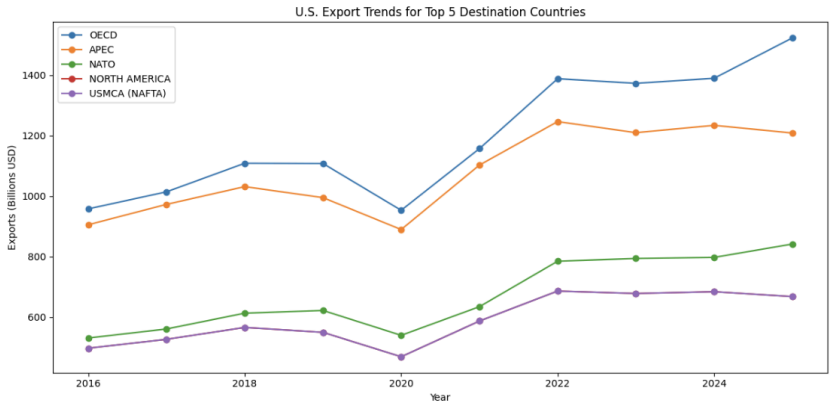
Leading Export Destinations and Geographic Stability

U.S. exports flow primarily to economically integrated and politically aligned blocs. The OECD and APEC remain the largest destinations, followed by the NATO alliance and regional partners in the USMCA (Fig. 5).



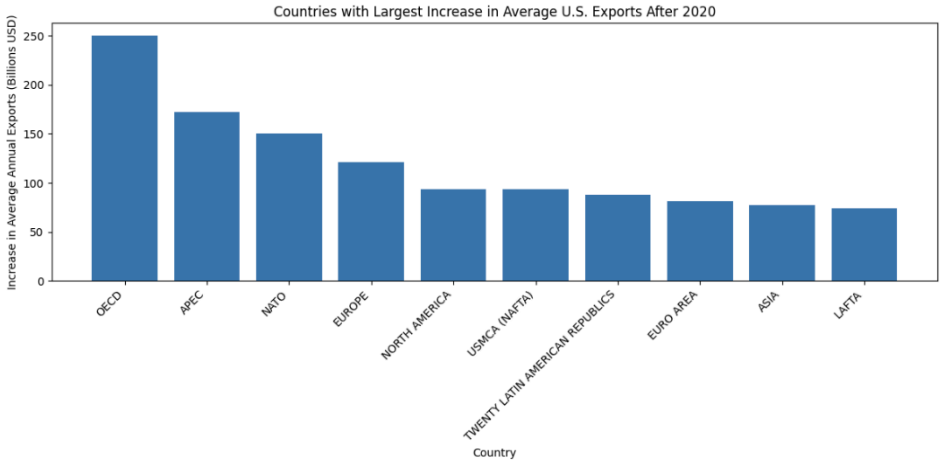
(**Fig. 5.** Top 10 U.S. export destination countries and regional groupings, 2016–2025)

The time-series data highlights a stability in trade where a bulk of trade is anchored in mature, high-income countries. Following the 2020 dip, the OECD and APEC regions showed the most aggressive "V-shaped" turns (Fig. 6).



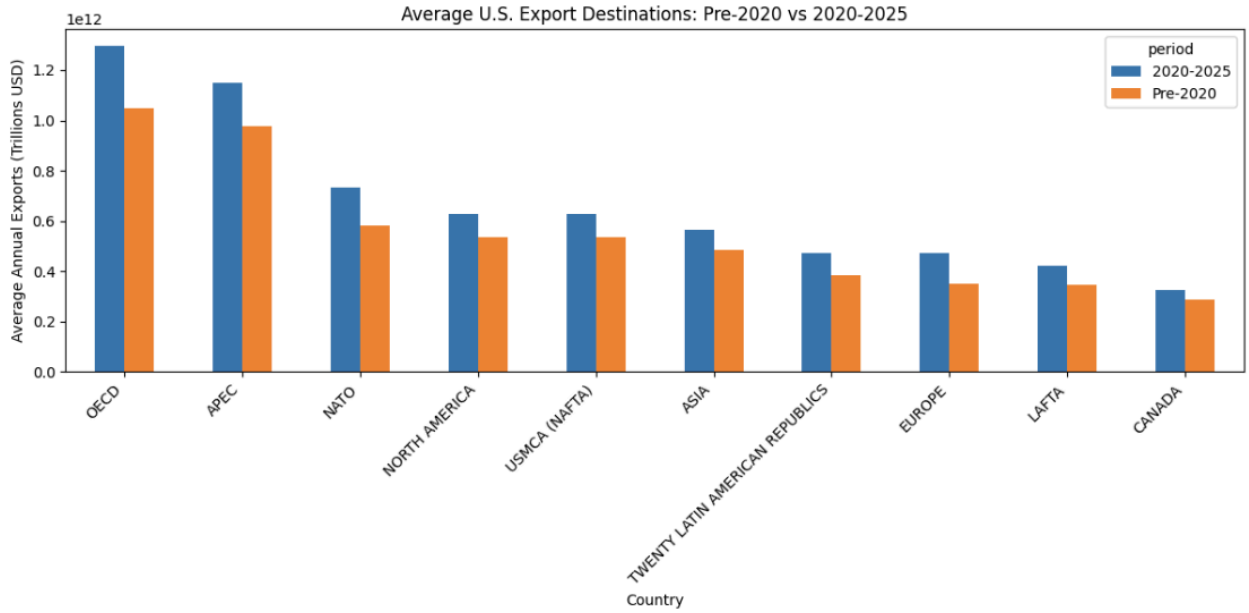
(**Fig. 6.** U.S. export trends for the top five destination countries or regional blocs, 2016–2025.)

When examining the "Largest Increases in Export Partners Post-2020," U.S. trade expansion has been deepening in existing ties rather than diversifying into new regions (Fig. 7).

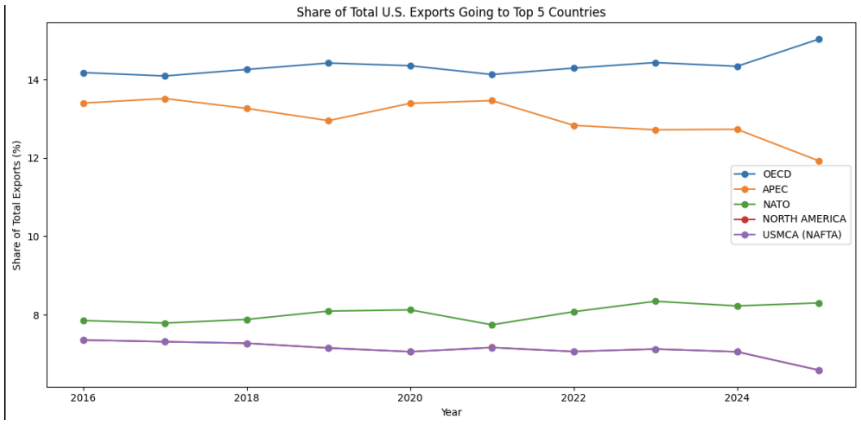


(**Fig. 7.** Countries and regional groupings with the largest increase in average U.S. exports after 2020)

This pattern also appears in the comparison between the pre-2020 and post-2020 periods, where the same major partners continue to dominate U.S. export flows (Figs. 8 & 9).



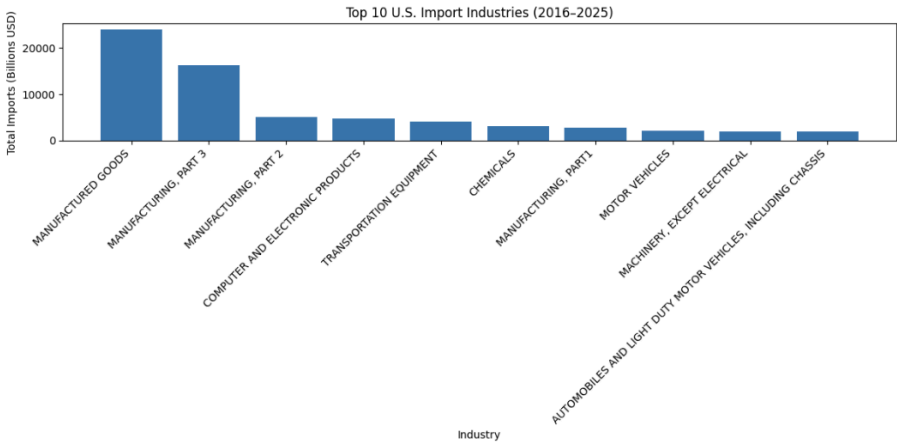
(**Fig. 8.** Average U.S. export destinations before 2020 and from 2020 to 2025)



(**Fig. 9.** Share of total U.S. exports going to the top five countries or regional blocs, 2016–2025)

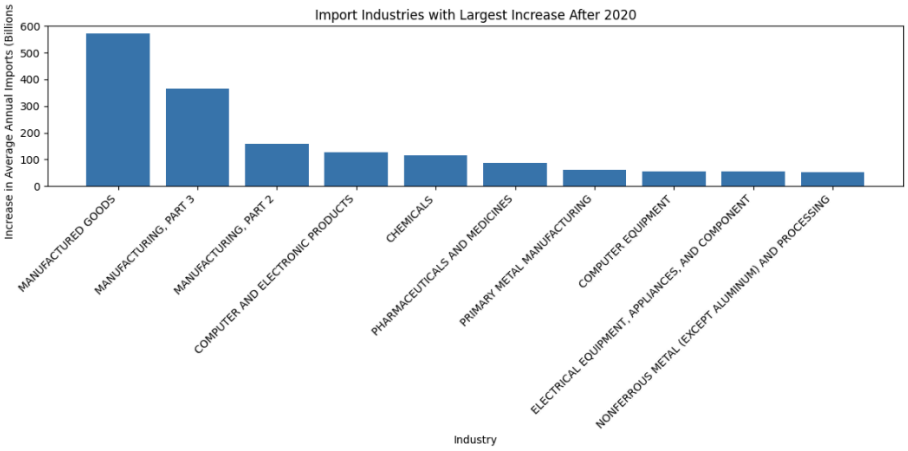
Top U.S. Import Industries and Post-2020 Shifts

U.S. imports exhibit an even higher concentration in Manufactured Goods than exports, with average annual values surging past the $2.5 trillion mark post-2020. Computer and Electronic Products and Chemicals also play major roles in the import ledger (Fig. 10).



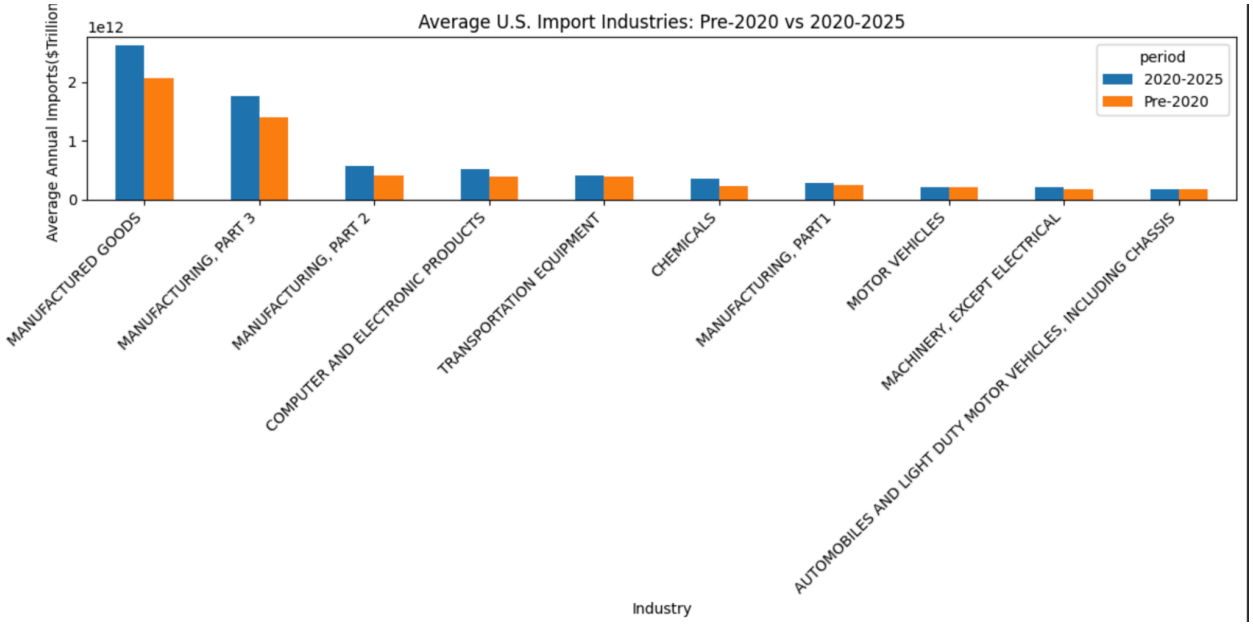
(**Fig. 10.** Top 10 U.S. import industries, 2016–2025)

A significant structural shift is visible in the post-2020 data. While manufactured goods saw the largest growth, categories like Oil and Gas and Chemicals/Pharmaceuticals experienced the most dramatic percentage increases in import value. We believe this reflects a strategic pivot in U.S. trade needs, likely driven by the dual pressures of securing pharmaceutical supply chains after the pandemic and navigating the global energy volatility that characterized 2022 and 2023 (Fig. 11).



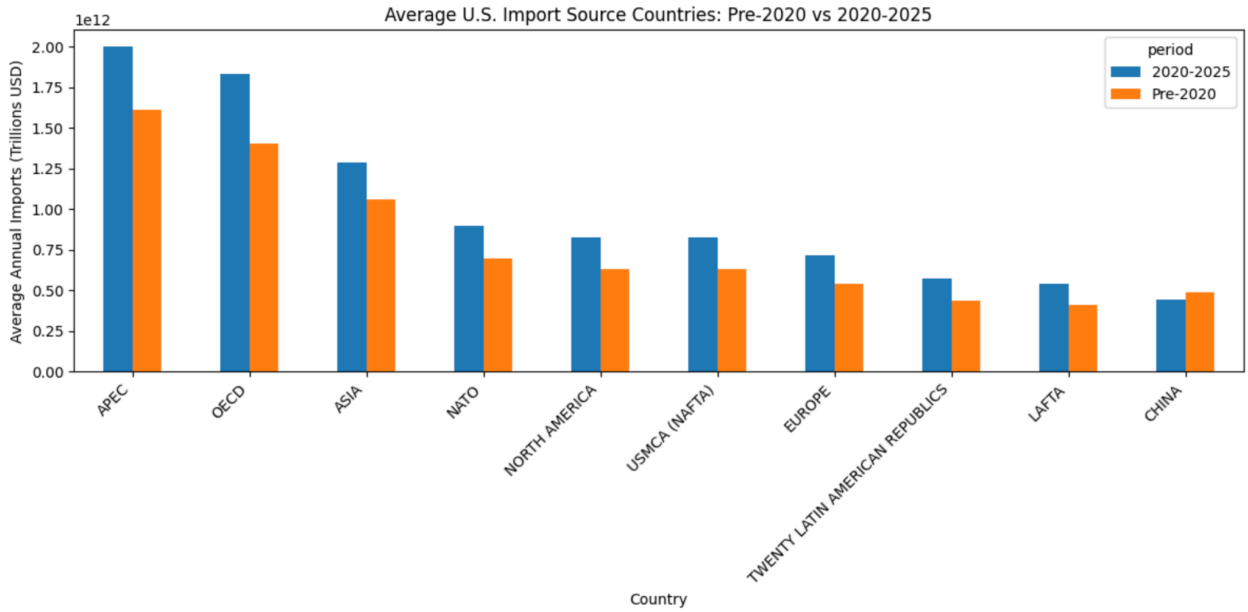
(**Fig. 11.** Import industries with the largest increase after 2020)

A comparison of pre-2020 and post-2020 averages confirms that most major import industries expanded in the later period (Fig. 12).



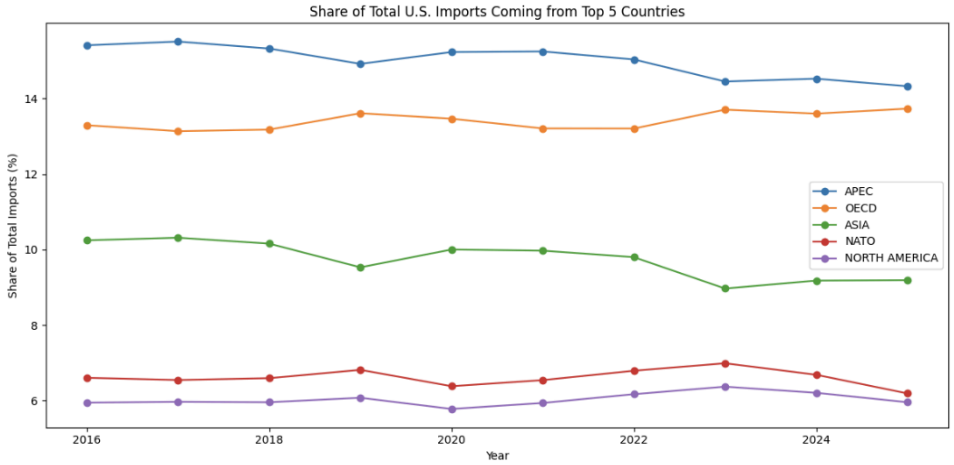
(**Fig. 12.** Average U.S. import industries before 2020 and from 2020 to 2025)

The distribution of U.S. import sources also changed after 2020, although the leading source regions remained concentrated among major economic blocs such as APEC, the OECD, and Asia **(**Fig. 13).



(**Fig. 13.** Average U.S. import source countries and regional groupings before 2020 and from 2020 to 2025)

The concentration of imports among the top five source partners also remained relatively stable over time, thus indicating continued dependence on a narrow set of major trade relationships (Fig. 14).



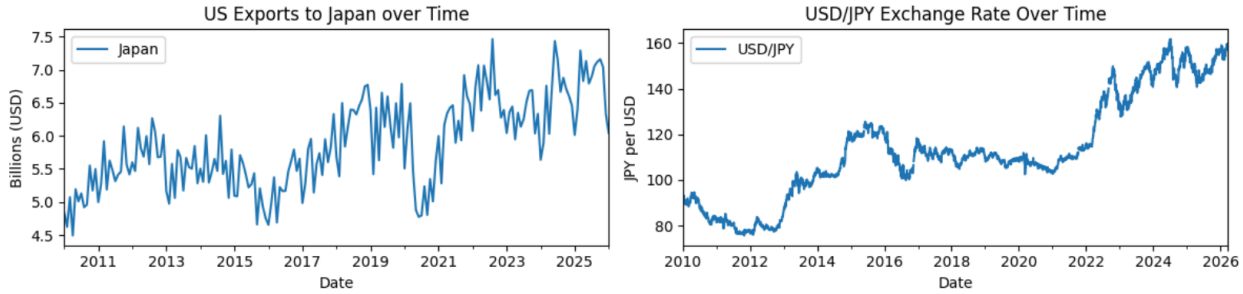
(**Fig. 14.** Share of total U.S. imports coming from the top five source countries or regional blocs, 2016–2025)

Analysis of Economic Factors on International Trade

Exchange Rates

The first factor we examined was exchange rates. An exchange rate is essentially how much of one currency is needed to purchase a unit of another (e.g. 1 US dollar would get you about 160 Japanese Yen). We hypothesized that a country with a higher exchange rate relative to another country would have higher imports and lower exports with that country. Our reasoning was that if you are able to purchase more of another country's currency with your own currency, then that nation's goods are cheaper for you to buy. Thus, you can import more (and vice versa with exports).

To test this hypothesis, we took a look at the change in US exchange rates over time and the corresponding US foreign exports. We used the exchange rates and exports to two nations: Japan and France (Japanese Yen and Euro respectively).



(**Fig. 15.** US Exports to Japan from 2010 to present & **Fig. 16.** USD/JPY Exchange Rate from 2010 to present)

For Japan, we can see the following trends. From 2010 to present, US exports to Japan have generally increased over time (Fig. 15). At the same time, the USD/JPY exchange rate has also generally increased over time (Fig. 16). This seems to suggest that the dollar strengthening relative to the yen over time hasn't resulted in less exports.

(**Fig. 17.** US Exports to France from 2010 to present & **Fig. 18.** USD/EUR Exchange Rate from 2010 to present)

Similarly for France, from 2010 to 2020, US exports to France increased slightly over time while the USD to EUR exchange rate shows no clear increasing or decreasing pattern to match. One exception is from 2021 to 2022 where US exports to France and the USD to EUR exchange rate seem to both increase at the same time (Figs. 17 and 18). However, that contradicts our initial hypothesis.

Overall, it seems that US exchange rates with other nations aren't a significant factor in determining US exports to other nations based on our observations. Across data from Japan and France, there was no clear pattern of behavior between US foreign exports and US exchange rates.

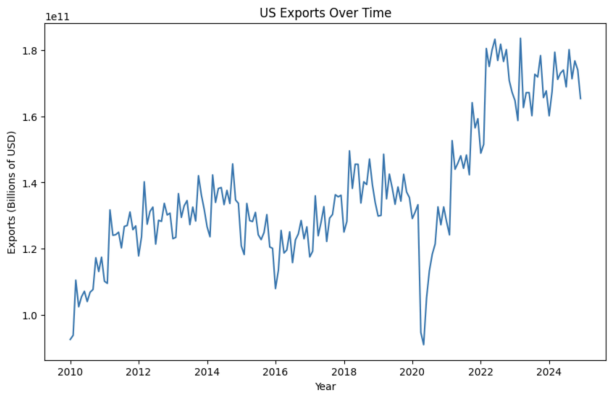
Energy Price (Oil Price)

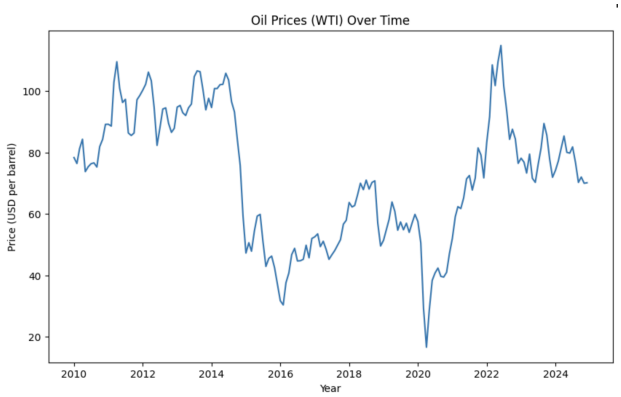
Apart from exchange rates, we looked at two other economic factors by not only visualizing the relevant data but also calculating the correlation between them and US export levels. For the second factor, we can see that the correlation between US exports and oil prices is 0.307, indicating a moderately positive relationship. This suggests that higher oil prices tend to coincide with somewhat higher US export values, though the link is relatively weak according to the number calculated.

This can potentially be explained through several mechanisms. Rising oil prices often signal strong global economic activity, boosting aggregate demand and international trade volume. Therefore, oil prices may act as an indirect indicator of global economic conditions rather than a direct driver of export performance.

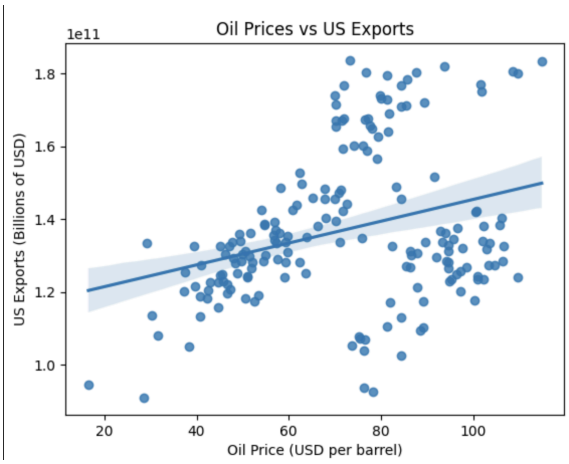
However, higher oil prices also raise transportation and production costs, potentially reducing export competitiveness. Since the US is a tremendous energy producer, rising oil prices simultaneously increase the value of energy-related exports, partially offsetting these costs. The moderate positive correlation

likely reflects this balance of opposing effects, suggesting oil prices are not the dominant factor influencing US trade performance.





(**Fig. 19.** US Exports from 2010 to 2025 & **Fig. 20.** Oil Prices from 2010 to 2025)



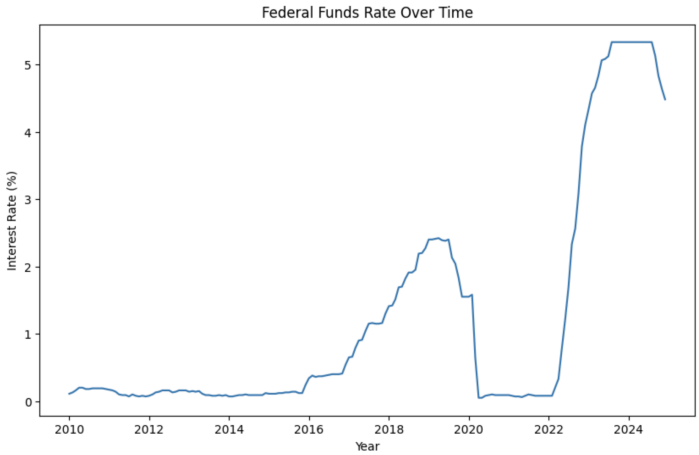
(**Fig. 21.** Oil Prices & US Export Linear Regression)

Monetary Policy (Interest Rates)

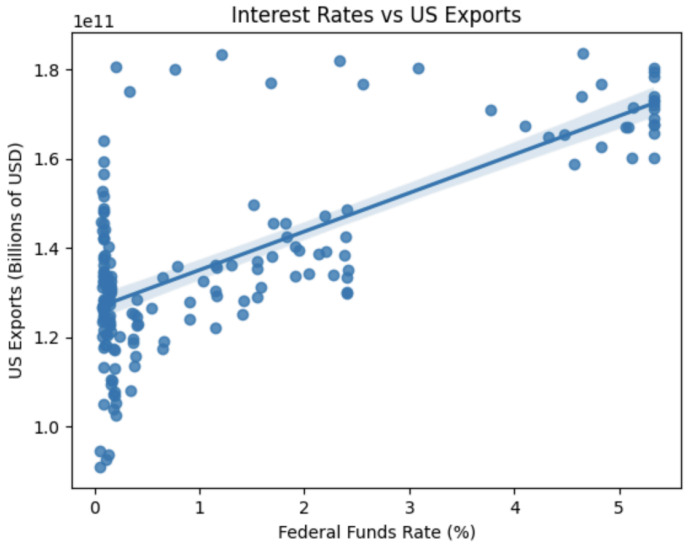
Finally, the correlation between US exports and the federal funds rate is calculated to be 0.711, indicating a relatively strong positive relationship (periods of higher interest rates tend to coincide with higher export levels).

This may seem counterintuitive, as higher interest rates typically slow economic activity by increasing borrowing costs, but this relationship likely reflects the broader macroeconomic environment rather than a direct causal effect. Central banks generally raise rates during periods of strong economic growth to curb inflation. These expansionary periods also drive higher production, investment, and international trade.

Moreover, higher interest rates attract global capital inflows, strengthening financial markets and supporting domestic economic activity, which in turn boosts US firms' capacity to supply international markets. The positive correlation therefore likely reflects both exports and interest rates rising together during robust economic growth, not a direct causal link.



(**Fig. 22.** Federal Funds Rate from 2010 to 2025)



(**Fig. 23.** Interest Rates & US Export Linear Regression)

Conclusion

In conclusion, firstly by looking at the overall trade landscape we've found that U.S. trade is overwhelmingly concentrated in manufactured goods and anchored in established economic blocs like OECD, APEC, and USMCA, a pattern that proved resilient even through the disruptions of 2020. Moreover, of the economic factors we examined, interest rates were the strongest correlating indicator ($r = 0.711$), followed by oil prices ($r = 0.307$), while exchange rates showed no meaningful relationship with export levels.

Surprisingly, U.S. trade consistently returned to the same partners and industries in the face of global shocks and policy shifts. This structural stability is the key takeaway of our investigation. Individual variables like currency fluctuations or oil prices may briefly affect trade flows, but we believe the deeper forces at work are the long-established relationships and economic cycles that quietly manipulate U.S. trade performance. This means that attempts to significantly change U.S. trade through tariffs, monetary policy, or currency actions are likely to be extremely difficult because existing patterns are so firmly established.

(Word Count excluding works cited and references is 1498)

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